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Semiconductor devices having exposed clip top sides and methods of manufacturing semiconductor devices

Abstract

A method includes providing a substrate having substrate terminals and providing a first component having a first terminal and a second terminal. The method includes providing a clip structure having a first clip, a second clip, and a clip connector coupling the first clip to the second clip. The method includes coupling the first clip to the first terminal and a substrate terminal and coupling the second clip to another substrate terminal. The method includes encapsulating the structure and removing a portion of the clip connector. In some examples, the first portion of the clip connector includes a first portion surface, the second portion of the clip connector includes a second portion surface, and the first portion surface and the second portion surface are exposed from a top side of the encapsulant. Other examples and related structures are also disclosed herein.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of co-pending U.S. patent application Ser. No. 18/108,590 and issued as U.S. Pat. No. 11,862,892 on Jan. 2, 2024, which is a divisional application of U.S. patent application Ser. No. 17/209,513 filed on Mar. 23, 2021 and issued as U.S. Pat. No. 11,611,170 on Mar. 21, 2023, which are incorporated by reference herein and priority thereto is hereby claimed.

TECHNICAL FIELD

(1) The present disclosure relates, in general, to electronic devices, and more particularly, to semiconductor devices and methods for manufacturing semiconductor devices.

BACKGROUND

(2) Prior semiconductor packages and methods for forming semiconductor packages are inadequate, for example resulting in excess cost, decreased reliability, relatively low performance, or package sizes that are too large. Further limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such approaches with the present disclosure and reference to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1B show cross-sectional views of an example semiconductor device and FIG. 1C shows a plan view of an example semiconductor device.
- (2) FIGS. 2A, 2B, 2C, 2D, 2E, 2F, and 2G show cross-sectional views and plan views of an example method for manufacturing an example semiconductor device.
- (3) FIGS. 3A and 3B show cross-sectional views of an example semiconductor device and FIG. 3C shows a plan view of an example semiconductor device.
- (4) FIGS. 4A, 4B, 4C, 4D, 4E, 4F, 4G, 4H, and 4I show cross-sectional views and plan views of an example method for manufacturing an example semiconductor device.
- (5) FIGS. 5A and 5B show plan view and cross-sectional view of an example method for manufacturing an example semiconductor device.
- (6) FIGS. 6A and 6B show plan view and cross-sectional view of an example method for manufacturing an example semiconductor device.
- (7) FIGS. 7A, 7B, and 7C show top view, side view and front view of an example method for manufacturing an example semiconductor device.
- (8) The following discussion provides various examples of semiconductor devices and methods of manufacturing semiconductor devices. Such examples are non-limiting, and the scope of the appended claims should not be limited to the particular examples disclosed. In the following discussion, the terms “example” and “e.g.” are non-limiting.
- (9) The figures illustrate the general manner of construction, and descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the present

disclosure. In addition, elements in the drawing figures are not necessarily drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of the examples discussed in the present disclosure. The same reference numerals in different figures denote the same elements.

(10) The term “or” means any one or more of the items in the list joined by “or”. As an example, “x or y” means any element of the three-element set $\{(x), (y), (x, y)\}$. As another example, “x, y, or z” means any element of the seven-element set $\{(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)\}$.

(11) The terms “comprises,” “comprising,” “includes,” or “including,” are “open ended” terms and specify the presence of stated features, but do not preclude the presence or addition of one or more other features.

(12) The terms “first,” “second,” etc. may be used herein to describe various elements, and these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, for example, a first element discussed in this disclosure could be termed a second element without departing from the teachings of the present disclosure.

(13) Unless specified otherwise, the term “coupled” May be used to describe two elements directly contacting each other or describe two elements indirectly connected by one or more other elements. For example, if element A is coupled to element B, then element A can be directly contacting element B or indirectly connected to element B by an intervening element C. Similarly, the terms “over” or “on” May be used to describe two elements directly contacting each other or describe two elements indirectly connected by one or more other elements.

DESCRIPTION

(14) In an example, a semiconductor device includes a substrate having substrate terminals, a first semiconductor component having a first component terminal and a second component terminal adjacent to a first major side of the first semiconductor component, and a clip structure including a first clip coupled to the first component terminal and a first substrate terminal, and a second clip coupled to a second substrate terminal. In some examples, an encapsulant covers the first semiconductor component, at least portions of the substrate, and at least portions of the clip structure. In some examples, a top side of the first clip and a top side of the second clip are exposed from a top side of the encapsulant.

(15) In an example, a semiconductor device includes a substrate having a first substrate terminal, a second substrate terminal, and a third substrate terminal. A semiconductor component includes a first major side, a second major side opposite to the first major side, a first component terminal and a second component terminal adjacent to the first major side, and the second major side can be coupled with the third substrate terminal. A clip structure includes a first clip having a first component-attached region with an upper surface, the first component-attached region coupled to the first component terminal, and a first substrate-attached region coupled to the first component-attached region and the first substrate terminal. The clip structure includes a second clip having a second component-attached region with an upper surface, and a second substrate-attached region coupled to the second component-attached region and the second substrate terminal. A first clip leg can be coupled to the first clip between the first clip and the second clip and the first clip leg can include a first leg end. A second clip leg can be coupled to the second clip between the first clip and the second clip, and the second clip leg can include a second leg end. an encapsulant can cover portions of the substrate, the first semiconductor component, and portions of the clip structure. In some examples, the first clip leg and the second clip leg are separated by a gap. In some examples, the first leg end and the second leg end are exposed from a major side of the encapsulant.

(16) In an example, a method of manufacturing a semiconductor device includes providing a substrate having substrate terminals and providing a first semiconductor component having a first component terminal and a second component terminal adjacent to a first major side of the first semiconductor component. The method includes providing a clip structure having a first clip, a second clip, and a clip connector coupling the first clip to the second clip. The method includes

coupling the first clip to the first component terminal and a first substrate terminal and coupling the second clip to a second substrate terminal. The method includes encapsulating the first semiconductor component, portions of the substrate, and portions of the clip structure. The method includes removing a sacrificial portion of the clip connector while leaving a first portion of the clip connector attached to the first clip and leaving a second portion of the clip connector attached to the second clip. The first portion of the clip connector includes a first portion surface, the second portion of the clip connector includes a second portion surface, and the first portion surface and the second portion surface are exposed from a top side of the encapsulant after the removing.

(17) Other examples are included in the present disclosure. Such examples may be found in the figures, in the claims, or in the description of the present disclosure.

(18) FIGS. 1A and 1B show cross-sectional views of an example semiconductor device **10**, and FIG. 1C shows a plan x-ray view of example semiconductor device **10**. FIG. 1A is a view taken from the perspective of line X-X of FIG. 1C, and FIG. 1B is a view taken from the perspective of line Y-Y of FIG. 1C. In the example shown in FIGS. 1A to 1C, semiconductor device **10** can comprise substrate **11**, semiconductor component **12**, encapsulant **13**, interface materials **14A**, **14B**, **14C**, **14D** and **14E**, and clip structure **15**.

(19) Substrate **11** can comprise substrate terminals **111**, **112** and **113**. Clip structure **15** can comprise clips **151** and **152**, clip joint **155** (illustrated in FIG. 2E), clip legs **1551** and **1552**. In some examples, clip joint **155** and clip legs **1551** and **1552** can be referred to as a clip connector, a conductive bridge, a conductive connector, a conductive bar, or a conductive interface.

(20) Substrate **11**, encapsulant **13** and clip structure **15** can be referred to as a semiconductor package and package can provide protection for semiconductor component **12** from external elements or environmental exposure. Semiconductor package can provide coupling between external electrical components and substrate terminals **111**, **112** and **113**.

(21) FIGS. 2A, 2B, 2C, 2D, 2E, 2F, and 2G show cross-sectional views and plan views of an example method for manufacturing semiconductor device **10**. In the following, reference is made to FIGS. 1A, 1B, and 1C together. FIG. 2A shows a cross-sectional view and plan view of semiconductor device **10** at an early stage of manufacture.

(22) In the example shown in FIG. 2A, substrate **11** can be provided. In some examples, substrate **11** can comprise or be referred to as a lead frame substrate, a laminate substrate, or a printed circuit board. In some examples, substrate **11** can comprise copper (Cu), Cu alloy, iron (Fe), Fe alloy or Fe—Ni alloy. In some examples, substrate **11** can comprise a coating or plating layer provided on a side of substrate **11**, such as nickel (Ni), palladium (Pd), gold (Au), silver (Ag), or solder (Sn).

Substrate **11** can comprise substrate terminals **111**, **112** and **113**. In some examples, substrate terminals **111**, **112** and **113** can comprise substantially rectangular plates. In some examples, substrate terminal **113** can have a larger area than substrate terminals **111** or **112**. In some examples, substrate terminals **111** or **112** can comprise or be referred to as leads. In some examples, substrate terminal **113** can comprise or be referred to as a lead, a paddle, a pad or a flag. In some examples, substrate terminals **111**, **112** and **113** can be provided through etching or stamping. In some examples, substrate **11** can have an area ranging from approximately 3 mm (millimeter)×3 mm to approximately 15 mm×15 mm. In some examples, substrate **11** can have a thickness ranging from approximately 100 μm (micrometer) to approximately 200 μm. In some examples, substrate terminals **111** or **112** can have an area ranging from approximately 1 mm×1 mm to approximately 10 mm×10 mm, or can have a thickness ranging from approximately 100 μm to approximately 200 μm. In some examples, an area of substrate terminal **113** can be dependent on a size of semiconductor component **12**, or can be in a range from approximately 1.5 mm×1.5 mm to approximately 15 mm×15 mm. In some examples, substrate terminal **113** can have a thickness ranging from approximately 100 μm to approximately 200 μm. Substrate **11** can serve as wiring for coupling semiconductor component **12** with an external component (for example, a motherboard or circuit board).

(23) FIG. 2B shows a cross-sectional view and plan view of semiconductor device **10** at a later stage of manufacture. In the example shown in FIG. 2B, interface material **14A** can be provided on substrate terminal **113**. In some examples, an area of interface material **14A** can be equal to or smaller than substrate terminal **113**. In some examples, interface material **14A** can comprise or be referred to as a solder, a conductive adhesive or a conductive paste. In some examples, interface material **14A** can comprise tin (Sn), silver (Ag), lead (Pb), copper (Cu), Sn—Pb, Sn37-Pb, Sn95-Pb, Sn—Pb—Ag, Sn—Cu, Sn—Ag, Sn—Au, Sn—Bi, or Sn—Ag—Cu. In some examples, interface material **14A** can be provided on substrate terminal **113** by coating or dispensing in the form of a paste. In some examples, interface material **14A** can have a thickness ranging from approximately 5 μm to approximately 100 μm . Interface material **14A** can couple semiconductor component **12** and substrate terminal **113**.

(24) FIG. 2C shows a cross-sectional view and plan view of semiconductor device **10** at a later stage of manufacture. In the example shown in FIG. 2C, semiconductor component **12** can be provided on substrate terminal **113**. In some examples, semiconductor component **12** can be arranged on interface material **14A**. Semiconductor component **12** can comprise or be referred to as a die, a chip, or a power component such as a field-effect transistor (FET) or an insulated-gate bipolar transistor (IGBT). In some examples, semiconductor component **12** can comprise component terminals **121** and **122** provided at a top side of semiconductor component **12**, and component terminal **123** provided at a bottom side of semiconductor component **12**. In some examples, component terminal **121** can comprise or be referred to as a source terminal (or a drain terminal). In some examples, component terminal **122** can comprise or be referred to as a gate terminal (or a control terminal). In some examples, component terminal **123** can comprise or be referred to as a drain terminal (or a source terminal). In some examples, electrical current can flow or can be prevented from flowing from the source terminal to the drain terminal (or vice versa) by a control signal supplied to the gate terminal. In some examples, semiconductor component **12** can have an area ranging from approximately 1 mm $\times$ 1 mm to approximately 10 mm $\times$ 10 mm. In some examples, semiconductor component **12** can have a thickness ranging from approximately 50 μm to approximately 775 μm . In some examples, component terminal **123** of semiconductor component **12** can contact interface material **14A**. In some examples, interface material **14A** can be supplied with heat and then cooled, and thus substrate terminal **113** can be coupled to component terminal **123** of semiconductor component **12** through interface material **14A**. In some examples, substrate **11** can be placed into a reflow furnace or a laser assist bonding apparatus, thereby applying approximately 150° C. to approximately 400° C. to interface material **14A**. Thereafter, cooling is performed, thereby coupling semiconductor component **12** to substrate terminal **113** through interface material **14A**.

(25) FIG. 2D shows a cross-sectional view and plan view of semiconductor device **10** at a later stage of manufacture. In the example shown in FIG. 2D, interface materials **14B** and **14C** can be provided on semiconductor component **12**. In some examples, interface materials **14B** and **14C** can be arranged on component terminals **121** and **122** of semiconductor component **12**, respectively. In some examples, interface materials **14D** and **14E** can be provided on substrate terminals **111** and **112**, respectively. In some examples, interface materials **14B**, **14C**, **14D** and **14E** can comprise or be referred to as a solder, a conductive adhesive or a conductive paste, or can be similar to interface material **14A**. In some examples, interface materials **14B**, **14C**, **14D** and **14E** can be provided on component terminals **121** and **122** and substrate terminals **111** and **112** by coating or by dispensing in the form of a paste. In some examples, interface materials **14B**, **14C**, **14D** and **14E** can have a thickness ranging from approximately 5 μm to approximately 100 μm . In some examples, interface materials **14B** and **14C** on component terminals **121** and **122** of semiconductor component **12**, and interface materials **14E** and **14D** on substrate terminals **111** and **112**, can couple clip structure **15** with substrate terminals **111** and **112** and with component terminals **121** and **122**.

(26) FIG. 2E shows a cross-sectional view and plan view of semiconductor device **10** at a later

stage of manufacture. In the example shown in FIG. 2E, clip structure **15** can be provided on component terminals **121** and **122** of semiconductor component **12**, and on substrate terminals **111** and **112** of substrate **11**. In some examples, clip structure **15** can comprise or be referred to as a conductive bridge, a conductive connector, a conductive bar, or a conductive interface. In some examples, clip structure **15** can comprise copper (Cu), Cu alloy, iron (Fe), Fe alloy, or Fe—Ni alloy. In some examples, clip structure **15** can comprise a coating layer or plating provided on a side of clip structure **15**, such as nickel (Ni), palladium (Pd), gold (Au), silver (Ag), or solder (Sn). In some examples, clip structure **15** can be formed using a material similar to that of substrate **11**, or can be formed in a similar manner as the substrate **11**. In some examples, clip structure **15** can be provided through etching or stamping.

(27) In some examples, clip structure **15** can comprise a substantially H shape in a plan view. In some examples, clip structure **15** can comprise clips **151** and **152** and clip joint **155**. In some examples, clip joint **155** can comprise clip legs **1551** and **1552** coupled to clips **151** and **152**, respectively. In some examples, clips **151** and **152** can be coupled to each other by clip joint **155**, clip leg **1551**, and clip leg **1552**. In some examples, clip joint **155** can have a greater height than clip **151**, clip **152**, clip leg **1551**, or clip leg **1552**. In some examples, opposite ends of clip joint **155** can be coupled to clips **151** and **152** through clip legs **1551** and **1552**, respectively. In some examples, clip legs **1551** and **1552** can be provided between clip joint **155** and clips **151** and **152** in an inclined shape that raises clip joint **155** relative to clips **151** and **152**. In some examples, because higher electrical current (e.g., source-drain current) can flow through clip **151**, and lower electrical current (e.g., a gate control signal) can flow through clip **152**, clip **151** can have a relatively larger width or area than clip **152**. In some examples, clip joint **155** can also be referred to as a sacrificial portion that is intended, for example, to be removed in a final semiconductor package. In other examples, clip **151** and clip **152** can have the same or equal widths or areas, for example if both clips are configured to have similar current load capabilities, or to meet other package design requirements.

(28) In some examples, one end of clip **151** can be coupled to semiconductor component **12**, and the other end of clip **151** can be coupled to substrate terminal **111**. In some examples, one end of clip **151** can be coupled to interface material **14B** provided on component terminal **121** of semiconductor component **12**, and the other end of clip **151** can be coupled to interface material **14D** provided on substrate terminal **111**. In some examples, clip **151** can be provided to have an inclined or stepped shape extending from semiconductor component **12** to substrate terminal **111**. In some examples, the one end of clip **151** coupled to semiconductor component **12** can be referred to as a component-attached region and the other end of clip **151** coupled to substrate terminal **111** can be referred to as a substrate-attached region.

(29) In some examples, one end of clip **152** can be coupled to semiconductor component **12**, and the other end of clip **152** can be coupled to substrate terminal **112**. In some examples, one end of clip **152** can be coupled to interface material **14C** provided on component terminal **122** of semiconductor component **12**, and the other end of clip **152** can be coupled to interface material **14E** provided on substrate terminal **112**. In some examples, clip **152** can be provided to have an inclined or stepped shape extending from semiconductor component **12** to substrate terminal **112**. In some examples, the one end of clip **152** coupled to semiconductor component **12** can be referred to as a component-attached region and the other end of clip **152** coupled to substrate terminal **112** can be referred to as a substrate-attached region.

(30) Specific shapes of clip structure **15** can be dependent on shapes or positions of substrate terminals **111**, **112** and **113**. In some examples, clip structure **15** can have a width ranging from approximately 200 μm to approximately 9500 μm , or a thickness ranging from approximately 100 μm to approximately 500 μm . In some examples, clip **151** or **152** can have a width ranging from approximately 200 μm to approximately 9500 μm , or a thickness ranging from approximately 100 μm to approximately 500 μm .

(31) In some examples, a top side of clip joint **155** can be approximately 300 μm to approximately 1300 μm higher than other regions excluding clip legs **1551** and **1552**. Clip joint **155** can have a thickness ranging from approximately 100 μm to approximately 500 μm . In some examples, clip legs **1551** and **1552** can have a width ranging from approximately 200 μm to approximately 1000 μm , and clip legs **1551** and **1552** can have a thickness ranging from approximately 100 μm to approximately 500 μm .

(32) In some examples, interface materials **14B**, **14C**, **14D** and **14E** can be supplied with heat and then cooled, and thus clip structure **15** can be coupled to substrate terminals **111** and **112** and component terminals **121** and **122** of semiconductor component **12** through interface materials **14B**, **14C**, **14D** and **14E**. In some examples, substrate **11** comprising clip structure **15** and interface materials **14B**, **14C**, **14D** and **14E** can be placed into a reflow furnace or a laser assist bonding apparatus, thereby applying a temperature of approximately 150° C. to approximately 400° C. to clip structure **15** and interface materials **14B**, **14C**, **14D** and **14E**. Thereafter, interface materials **14B**, **14C**, **14D** and **14E** can be cooled, and thus clip structure **15** and semiconductor component **12** can be coupled to each other through interface materials **14B**, **14C**, **14D** and **14E**.

(33) In some examples, a melting point of interface material **14A** between semiconductor component **12** and substrate terminal **113** can be higher than interface materials **14B**, **14C**, **14D** and **14E** between semiconductor component **12** and clip structure **15** or clip structure **15** and substrate terminals **111** and **112**. When clip structure **15** is coupled to semiconductor component **12** and substrate terminals **111** and **112** through interface materials **14B**, **14C**, **14D** and **14E**, interface materials **14B**, **14C**, **14D** and **14E**, except for interface material **14A**, can be melted. Accordingly, while clip structure **15** is coupled to semiconductor component **12** and substrate terminals **111** and **112**, semiconductor component **12** can be prevented from rotating or shifting on substrate terminal **113**. Adjusting of such melting point can be achieved by adjusting the content of solders in interface materials or varying kinds or composition ratios of alloys.

(34) FIG. 2F shows a cross-sectional view and x-ray plan view of semiconductor device **10** at a later stage of manufacture. In the example shown in FIG. 2F, encapsulant **13** can be provided on substrate **11**, semiconductor component **12** and clip structure **15**. Encapsulant **13** can contact substrate **11**, semiconductor component **12** and clip structure **15**, or can encapsulate substrate **11**, semiconductor component **12** and clip structure **15**. As seen in FIG. 2F, encapsulant **13** can be applied to fully cover clip structure **15**. There can be examples where a portion of clip structure **15**, such as clip joint **155**, can remain exposed from encapsulant **13**. In some examples, a region of substrate **11** can be exposed through encapsulant **13**. In some examples, bottom sides of substrate terminals **111**, **112** and **113** can be exposed at a bottom side of encapsulant **13**. In some examples, the bottom side of encapsulant **13** can be coplanar with the bottom sides of substrate terminals **111**, **112** and **113**.

(35) Encapsulant **13** can comprise or be referred to as a mold compound, a resin, a sealant, a filler-reinforced polymer, or a package body. In some examples, encapsulant **13** can comprise an epoxy or phenol resin, carbon black and a silica filler. In some examples, encapsulant **13** can be provided by compression molding, transfer molding, liquid encapsulant molding, vacuum lamination, paste printing or film assist molding. The compression molding can be performed by supplying a flowable resin to a mold in advance, placing a substrate into the mold and then curing the flowable resin, and the transfer molding can be performed by supplying a flowable resin to a gate (supply port) of a mold and to surroundings of a pertinent substrate and then curing the flowable resin. Encapsulant **13** can have a width ranging from approximately 3 mm×3 mm to approximately 15 mm×15 mm, and a thickness ranging from approximately 0.7 mm to approximately 2.1 mm. Encapsulant **13** can provide protection for a semiconductor component from external elements or environmental exposure and can rapidly emit heat generated from the semiconductor component outward.

(36) FIG. 2G shows a cross-sectional view and x-ray plan view of semiconductor device **10** at a

later stage of manufacture. In the example shown in FIG. 2G, encapsulant **13** can be thinned, for example, by grinding with a grinder. In some examples, a top side of encapsulant **13** can be thinned until clip joint **155** is removed to disconnect clip **151** and clip **152** from each other. In some examples, clip joint **155** can be grinded or removed while encapsulant **13** is thinned. In some examples, clip joint **155** can be cut with a mechanical or laser saw. With clip joint **155** removed, clips **151** and **152** are physically and electrically disconnected from each other. In some examples, source-drain current can then independently flow through clip **151**, and a gate control signal can be independently transmitted through clip **152**. As illustrated in FIG. 2G, after clip joint **155** is removed, a gap **155A** or space **155A** is interposed between clip leg **1552** and clip leg **1551**.

(37) In some examples, after the grinding, top sides of clip legs **1551** and **1552** can remain exposed at the top side of encapsulant **13**. In some examples, the grinding can continue until clip legs **1551** and **1552** are also removed. In some examples, after the grinding, top sides of clips **151** and **152** can be exposed at the top side of encapsulant **13**. In some examples, the grinding or removal of clip joint **155** can comprise a stage, features or elements similar to those described with respect to FIG. 7A-7C.

(38) In some examples, the stages above can be followed by performing general plating, marking, singulating and shipping. In some examples, the plating can comprise supplying an oxidation resistant film to clip legs **1551** and **1552**, clips **151** and **152**, or substrate terminals **111**, **112** and **113**, exposed from the top side or the bottom side of encapsulant **13**. In some examples, the oxidation resistant film can comprise gold (Au), silver (Ag), nickel (Ni), palladium (Pd), solder (Sn), or organic solderability preservative (OSP). In some examples, active elements, such as a semiconductor die, an electronic component, or passive elements such as an inductor or a capacitor, can be mounted on clip legs **1551** and **1552** or clips **151** and **152** exposed from encapsulant **13**. The marking can comprise marking a product name or a manufacturer's name on a side of encapsulant **13**. The singulating can comprise separating semiconductor devices fabricated in a matrix or stripe configuration having multiple rows or columns into individual semiconductor devices by sawing/cutting. The shipping can comprise placing the individual semiconductor devices into an antistatic tray.

(39) According to the present disclosure, even if semiconductor device **10** or clip structure **15** is small or narrow, clip structure **15** on semiconductor component **12** can be prevented from falling over or shifting during the manufacture of semiconductor device **10** because of the stability provided by clip joint **155** tying or coupling clips **151** and **152** together. In some examples, during the manufacture of semiconductor device **10**, clip structure **15** having a substantially H-shaped configuration can be provided, and clip structure **15** can be divided into individual clips by grinding or grooving after the encapsulating, thereby providing clip structure **15** at accurate positions between semiconductor component **12** and substrate terminals **111** and **112**. In some examples, active elements or passive elements can be mounted on clips **151** and **152** exposed through encapsulant **13**, and thus application ranges of semiconductor device **10** can be extended.

(40) FIGS. 3A and 3B show cross-sectional views of an example semiconductor device **20** and FIG. 3C shows an x-ray plan view of an example semiconductor device **20**. FIG. 3A is a cross-sectional view taken along line Y-Y of FIG. 3C, and FIG. 3B is a cross-sectional view taken along line X-X of FIG. 3C. In the example shown in FIGS. 3A to 3C, semiconductor device **20** can comprise substrate **21**, semiconductor component **12** and **22**, encapsulant **13**, interface materials **14A**, **14B**, **14C** and **14D**, clip structure **15** and interconnect **29**. Substrate **21** can comprise substrate terminals **211**, **213** and **214**. Semiconductor device **20** can be similar to the above-described semiconductor device **10** terms of features, elements, or manufacturing. As illustrated in FIG. 3C, after clip joint **155** is removed, gap **155A** is interposed between clip leg **1552** and clip leg **1551**.

(41) FIGS. 4A, 4B, 4C, 4D, 4E, 4F, 4G, 4H, and 4I show cross-sectional views and plan views of an example method for manufacturing semiconductor device **20**. In the following, reference is made to FIGS. 3A, 3B, and 3C together.

(42) FIG. 4A shows a cross-sectional view of semiconductor device **20** at an early stage of manufacture. In the example shown in FIG. 4A, substrate **21** can be provided. Interface materials **14A** can be provided on substrate terminals **113** and **214**, similar to as described with respect to interface material **14A** for FIG. 2B.

(43) In some examples, substrate **21** can be similar to substrate **11**. Substrate **21** can comprise substrate terminals **211**, **113** and **214**. Substrate terminals **211** can be arranged at peripheral edges of substrate terminals **113** and **214**. In some examples, multiple substrate terminals **211** can be arranged at one side of a substrate terminal **113**. In some examples, multiple substrate terminals **211** can be arranged at three side sides of substrate terminals **113** or **214**. In some examples, substrate terminal **211** can comprise or be referred to as one or more leads. In some examples, substrate terminals **113** or **214** can comprise or be referred to a leads, pads, paddles or flags.

(44) In some examples, substrate **21** can have a width ranging from approximately 3 mm×3 mm to approximately 15 mm×15 mm. In some examples, substrate **21** can have a thickness ranging from approximately 100 μm to approximately 200 μm. In some examples, substrate terminals **211** can have a width ranging from approximately 1 mm×1 mm to approximately 10 mm×10 mm. In some examples, substrate terminals **211** can have a thickness ranging from 100 μm to approximately 200 μm. An area of substrate terminal **214** can be dependent on the size of semiconductor component **22**, and in some examples substrate terminal **214** can have an area ranging from approximately 1.5 mm×1.5 mm to approximately 10.5 mm×10.5 mm, or a thickness ranging from approximately 100 μm to approximately 200 μm.

(45) FIG. 4B shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. 4B, semiconductor component **12** can be provided on substrate terminal **113**. In some examples, the stage, features, or elements of FIG. 4B can be similar to those described with respect to semiconductor component **12** for FIG. 2C. In some examples, semiconductor component **12** can be arranged on interface material **14A**. Semiconductor component **12** can comprise component terminals **121** and **122** at a top side of semiconductor component **12**, and component terminal **123** at a bottom side of semiconductor component **12**.

(46) In some examples, semiconductor component **22** can be provided on substrate terminal **214**. In some examples, semiconductor component **22** can be arranged on interface material **14A**. In some examples, semiconductor component **22** can comprise or be referred to as a controller, a digital signal processor (DSP), a microprocessor, a network processor, a power management processor, an audio processor, an RF circuit, a wireless baseband system-on-chip (SoC) processor, a sensor, or an application specific integrated circuit (ASIC). In some examples, semiconductor component **22** can have an area ranging from approximately 1 mm×1 mm to approximately 10 mm×10 mm. In some examples, semiconductor component **22** can have a thickness ranging from 50 μm to approximately 775 μm. In some examples, semiconductor component **22** can comprise multiple terminals **221** located on a top side of semiconductor component **22**. In some examples, interface material **14A** is melted or cured to couple semiconductor components **12** and **22** to substrate terminals **113** and **214** respectively.

(47) FIG. 4C shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. 4C, interface materials **14B**, **14C**, and **14D** can be provided. In some examples, interface materials **14B**, **14C**, or **14D** can be applied similar to as described with respect to FIG. 2D. In some examples, interface material **14B** and **14C** can be respectively provided on component terminal **121** and **122** of semiconductor component **12**. In some examples, interface material **14D** can be provided on substrate terminals **211** of substrate **21**.

(48) FIG. 4D shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. 4D, clip structure **15** can be provided on semiconductor component **12** and substrate terminal **211**. In some examples, the stage, features, or elements of FIG. 4D can be similar to those described with respect to clip structure **15** for FIG. 2E.

(49) In some examples, clip structure **15** can comprise clips **151** and **152** (see FIG. 3C) and clip

joint **155**. In some examples, clip joint **155** can comprise clip legs **1551** and **1552** (see FIG. 3C). Clips **151** and **152** can be coupled to each other by clip joint **155**. In some examples, clip joint **155** can have a greater height than clips **151** and **152**. In some examples, opposite ends of clip joint **155** can be coupled to clips **151** and **152** through clip legs **1551** and **1552**, respectively. In some examples, clip legs **1551** and **1552** can be provided between clip joint **155** and clips **151** and **152** in an inclined shape.

(50) In some examples, first ends of clips **151** and **152** can be coupled to semiconductor component **12**, and second ends of clips **151** and **152** can be coupled to substrate terminal(s) **211**. In some examples, the first ends of clips **151** and **152** can be coupled to interface material **14B** provided on component terminal **121** of semiconductor component **12**, and the second ends of clips **151** and **152** can be coupled to interface material **14D** provided on substrate terminal **211**. In some examples, clips **151** and **152** can be provided in an inclined or stepped shape extending from semiconductor component **12** to substrate terminal(s) **211**. In some examples, the first ends of clips **151** and **152** can be referred to as component-attached regions. In some examples, the second ends of clips **151** and **152** can be referred to as substrate-attached regions.

(51) FIG. 4E shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. 4E, a reflow process can be performed. The reflow process can comprise placing substrate **21** into a reflow furnace or a laser assist bonding device to apply a temperature of approximately 150° C. to approximately 400° C. Thereafter, substrate **21** can be cooled and thus melted interface materials **14B** and **14D** can be cooled, thereby coupling substrate terminal **211** and component terminal **121** of semiconductor component **12** to each other through clip structure **15**.

(52) FIG. 4F shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. 4F, cleaning can be performed. The cleaning can comprise removing residuals of interface materials or removing a variety of particles remaining on substrate **11**, semiconductor component **12** and **22**, and clip structure **15**. In some examples, the cleaning can comprise a variety of processes including, for example, spraying a washing solution onto substrate **11**, soaking substrate **21** into a washing solution tank, or ultrasonically washing.

(53) FIG. 4G shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. 4G, interconnects **29** can be provided. In some examples, semiconductor component **12** and semiconductor component **22** can be coupled to each other by interconnect **29**, and semiconductor component **22** and substrate terminal **211** can be coupled to each other by interconnect **29**. In some examples, terminal **221** of semiconductor component **22** and component terminal **122** (e.g., a gate terminal) of semiconductor component **12** can be bonded to each other by interconnect **29**. In some examples, terminal **221** of semiconductor component **22** and substrate terminal **211** can be bonded to each other by interconnect **29**. In some examples, a first end of interconnect **29** can be ball-bonded to terminal **221** of semiconductor component **22**, and a second end of interconnect **29** can be stitch-bonded to component terminal **122** of semiconductor component **12**, or vice versa. In some examples, the first end of interconnect **29** can be ball-bonded to terminal **221** of semiconductor component **22**, and the second end of interconnect **29** can be stitch-bonded to substrate terminal **211**, and vice versa. In some examples, interconnect **29** can comprise or be referred to as a conductive wire or a bonding wire. In some examples, interconnect **29** can have a diameter ranging from approximately 15 μm to approximately 30 μm . Interconnect **29** can transfer an electrical signal (e.g., a control signal) from semiconductor component **22** to semiconductor component **12**.

(54) FIG. 4H shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. 4H, encapsulant **13** can be provided on substrate **21**, semiconductor component **12** and **22**, and clip structure **15**. In some examples, the stage, features, or elements of FIG. 4H can be similar to those described with respect to encapsulant **13** for FIG. 2F. Encapsulant **13** can contact substrate **21**, semiconductor component **12** and **22**, and clip

structure **15**, or can encapsulate substrate **21**, semiconductor component **12** and **22**, and clip structure **15**. As seen in FIG. **4H**, encapsulant **13** can be applied to fully cover clip structure **15**. There can be examples where a portion of clip structure **15**, such as clip joint **155**, can remain exposed from encapsulant **13**.

(55) FIG. **4I** shows a cross-sectional view of semiconductor device **20** at a later stage of manufacture. In the example shown in FIG. **4I**, thinning can be performed. In some examples, the stage, features, or elements of FIG. **4I** can be similar to those described with respect to thinning for FIG. **2G**. In some examples, a top side of encapsulant **13** can be thinned until clip joint **155** is removed to disconnect clip **151** and clip **152** from each other. In some examples, clip joint **155** can be grinded or removed while encapsulant **13** is thinned. In some examples, clip joint **155** can be cut with a mechanical or laser saw. With clip joint **155** removed, clips **151** and **152** are physically and electrically disconnected from each other. In some examples, source-drain current can then independently flow through clip **151**, and source-drain current can then independently flow through clip **152**.

(56) FIGS. **5A** to **5B** show plan view and cross-sectional view of an example method for manufacturing example semiconductor device **20**. In the example shown in FIGS. **5A** and **5B**, after the grinding, clip legs **1551** and **1552** can be exposed at the top side of encapsulant **13**. In some examples, an oxidation resistant film made of gold (Au), silver (Ag), nickel (Ni), palladium (Pd), solder (Sn), or organic solderability preservative (OSP) can be provided on clip legs **1551** and **1552**. In some examples, component **90** can be mounted on clip legs **1551** and **1552**. In some examples component **90** can comprise active elements, such as a semiconductor die, or passive elements such as an inductor or a capacitor.

(57) FIGS. **6A** to **6B** show plan view and cross-sectional view of an example method for manufacturing example semiconductor device **20**. In the example shown in FIGS. **6A** and **6B**, after the grinding, clips **151** and **152** or clip legs **1551** and **1552** can be exposed at the top side of encapsulant **13**. Note that in some examples the grinding can continue until clip legs **1551** and **1552** are completely removed, such that only clips **151** and **152** are exposed at the top side of encapsulant **13**. In some examples, an oxidation resistant film made of gold (Au), silver (Ag), nickel (Ni), palladium (Pd), solder (Sn), or organic solderability preservative (OSP) can be provided on clips **151** and **152** or clip legs **1551** and **1552**. In some examples, component **90** can be mounted on clip legs **1551** and **1552** or clips **151** and **152**. In some examples component **90** can comprise active elements, such as a semiconductor die, electronic components, or passive elements such as an inductor or a capacitor.

(58) FIGS. **7A** to **7C** show top view, side view and front view of portions of example methods for manufacturing semiconductor device **10** or **20**.

(59) In some examples, as illustrated in FIG. **7A**, clip structure **15** can comprise clips **151** and **152** positioned adjacent each other, with clip joint **155** positioned between clips **151** and **152**, with clip leg **1551** coupling clip **151** with clip joint **155**, and with clip leg **1552** coupling the clip **152** with clip joint **155**. Clip legs **1551** and **1552** can extend inclined between clip joint **155** and respective clips **151-152**. A top of clip joint **155** can be higher than the top of clips **151** and **152**, or higher than midpoints of clip legs **1551** and **1552**.

(60) In some examples, clips **151** and **152** can also comprise inclined portions. For instance, clip **151** can comprise substrate-attached region **151A** positioned lowest coupled to substrate terminal **111** or **211**, component-attached region **151B** positioned highest coupled to semiconductor component **12**, and inclined region **151C** coupling substrate-attached region **151A** with component-attached region **151B**.

(61) After encapsulating with encapsulant **13**, clip joint **155** can be removed. In some examples, such removal of clip joint **155** can correspond or be similar to the stages described with respect to FIG. **2F-2G**, **4H-4I**, **5A-5B**, or **6A-6B**. In some examples clip joint **155** can be grinded to disconnect clips **151** and **152** from each other, with clip legs **1551** and **1552** or clips **151** and **152**

remaining exposed at the top side of encapsulant **13**. In some examples, component **90** can be mounted on exposed clip legs **1551** and **1552** or exposed clips **151** and **152**.

(62) In some examples, as shown in FIG. 7B, clip structure **15** can comprise clips **151** and **152** positioned adjacent each other, with clip joint **155** positioned between clips **151** and **152**. Clip legs **1551** and **1552** can be optional in the present example or can be considered part of clip joint **155**. In the present example, without clip legs **1551** and **1552** inclined, the top of clip joint **155** can be coplanar with or at similar height as the top of clips **151** or **152**.

(63) In some examples, clip joint **155** can initially tie together component-attached regions **151B** and **152B** of clips **151** and **152**. In some examples, clip joint **155** can be substantially coplanar with component attached region **151B**. In some examples, clip joint **155** can comprise a thin region or groove **1553** at the bottom of clip joint **155**.

(64) After encapsulating with encapsulant **13**, clip joint **155** can be removed. In some examples, such removal of clip joint **155** can correspond or be similar to the stages described with respect to FIG. 2F-2G, 4H-4I, 5A-5B, or 6A-6B. In some examples clip joint **155** can be grinded until removed, such as by grinding to reach or expose groove **1553**, thus disconnecting clips **151** and **152** from each other, with clips **151** and **152** remaining exposed at the top side of encapsulant **13**. In some examples, component **90** can be mounted on exposed clips **151** and **152**.

(65) In some examples, as illustrated in FIG. 7C, clip structure **15** can comprise clips **151** and **152** positioned adjacent each other, with clip joint **155** positioned between clips **151** and **152**. Clip legs **1551** and **1552** can be optional in the present example or can be considered part of clip joint **155**. In the present example, without clip legs **1551** and **1552** inclined, the top of clip joint **155** can be coplanar with or at similar height as the top of clips **151** or **152**. In some examples, clip structure **15** of FIG. 7C can initially be similar to the clip structure of FIG. 7A or 7B.

(66) After encapsulating with encapsulant **13**, clip joint **155** can be removed. In some examples, such removal of clip joint **155** can correspond or be similar to the stages described with respect to FIG. 2F-2G, 4H-4I, 5A-5B, or 6A-6B. In some examples, after the encapsulating with encapsulant **13**, clip joint **155** can be removed by partial cutting, such as by sawing with a mechanical or laser saw, whether alone or after grinding. In some examples, clip joint **155** can be cut, thereby separating clips **151** and **152** from each other. In some examples, the partial cutting can be performed linearly across encapsulant **13** and clip joint **155** using a diamond wheel or a laser beam. In some examples, because of the partial cutting, encapsulant **13** can comprise groove **131**.

(67) In some examples, partial cutting to remove clip joint **155** can be carried out after grinding to expose clip joint **155**. In some examples, during the encapsulation with encapsulant **13**, the top of clip structure **15** or clip joint **155** remain be exposed. In some examples, encapsulant **13** can encapsulate a lower region of clip structure **15**, except for the top of clip structure **15** or clip joint **155**, by film assist molding. After encapsulating, the top side of clip structure **15** can remain exposed even without grinding, and grinding can be omitted. Partial cutting can be performed on the exposed clip joint **155**, thereby untying separating clip structure **15** into two clips **151** and **152** from each other. FIGS. 6A, 7A, and 7B show examples where upper surfaces of component-attached regions (e.g., **151B**, **152B**) of clip structure **15** are exposed from a top side of encapsulant **13**.

(68) The present disclosure includes reference to certain examples; however, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the disclosure. In addition, modifications may be made to the disclosed examples without departing from the scope of the present disclosure. Therefore, it is intended that the present disclosure not be limited to the examples disclosed, but that the disclosure will include all examples falling within the scope of the appended claims.

Claims

1. A method of manufacturing an electronic device, comprising: coupling a first electronic component to a first side of a substrate, the first electronic component comprising a first component terminal and a second component terminal located on a first major side of the first electronic component; providing a clip structure comprising: a first clip; a second clip; and a clip connector coupled to the first clip and the second clip; coupling the first clip to the first component terminal; coupling the second clip to a first terminal of the substrate; providing an encapsulant over the first electronic component, portions of the substrate, and portions of the clip structure; and removing at least a portion of the clip connector to separate the first clip from the second clip, wherein a first top side of the first clip and a second top side of the second clip are exposed from a top side of the encapsulant.
2. The method of claim 1, further comprising: providing a second electronic component comprising a first component terminal and a second component terminal located on a first major side of the second electronic component; wherein: coupling the second clip comprises coupling the second clip to the first component terminal of the second electronic component.
3. The method of claim 1, wherein: coupling the first clip comprises coupling the first clip to a second terminal of the substrate.
4. The method of claim 1, wherein: removing comprises leaving a first portion of the clip connector attached to the first clip and leaving a second portion of the clip connector attached to the second clip; and the first portion of the clip connector and the second portion of the clip connector are exposed from the encapsulant.
5. The method of claim 1, wherein: coupling the second clip comprises coupling the second clip to the second component terminal of the first electronic component.
6. The method of claim 1, wherein: removing comprises removing a portion of the encapsulant before removing at least the portion of the clip connector.
7. A method of manufacturing an electronic device, comprising: coupling a first electronic component to a substrate; providing a clip structure comprising: a first clip; a second clip; and a clip connector coupled to the first clip and the second clip; coupling the first clip to a first component terminal of the first electronic component; coupling the second clip to a first terminal of the substrate; providing an encapsulant over the first electronic component, the substrate, and the clip structure; and removing at least a portion of the clip connector to separate the first clip from the second clip and expose the first clip and the second clip from a top side of the encapsulant.
8. The method of claim 7, wherein: removing comprises leaving a first portion of the clip connector attached to the first clip and leaving a second portion of the clip connector attached to the second clip; and the first portion of the clip connector and the second portion of the clip connector are exposed from the encapsulant.
9. The method of claim 7, wherein: coupling the first clip comprises coupling the first clip to a second terminal of the substrate.
10. The method of claim 7, further comprising: coupling a second electronic component to the substrate, the second electronic component comprising a first component terminal and a second component terminal located on a first major side of the second electronic component; and coupling the first component terminal of the second electronic component to a second component terminal of the first electronic component.
11. The method of claim 10, further comprising: coupling a third electronic component to the substrate, the third electronic component comprising a first component terminal and a second component terminal located on a first major side of the third electronic component; wherein: providing the clip structure comprises: providing the first clip coupled to a second substrate terminal; and providing the second clip coupled to the first component terminal of the third electronic component.
12. The method of claim 7, further comprising: coupling a component to the top side of the first

clip and the top side of the second clip.

13. The method of claim 7, wherein providing the clip structure comprises providing: a first clip leg extending from the first clip towards the second clip; a second clip leg extending from the second clip towards the first clip, the first clip comprising a first attachment region having an upper surface; and the second clip comprising a second attachment region having an upper surface.

14. The method of claim 13, wherein: providing the clip structure comprises providing the first clip leg and the second clip leg extending upward such that an upper surface of the first clip leg is above the upper surface of the first attachment region and an upper surface of the second clip leg is above the upper surface of the second attachment region; the portion of the first clip exposed from the top side of the encapsulant comprises the upper surface of the first clip leg; and the portion of the second clip exposed from the top side of the encapsulant comprises the upper surface of the second clip leg.

15. The method of claim 13, wherein: providing the clip structure comprises: providing an upper surface of the first clip leg and the upper surface of the first attachment region coplanar so that the portion of the first clip exposed from the top side of the encapsulant comprises the upper surface of the first attachment region; and providing an upper surface of the second clip leg and the upper surface of the second attachment region coplanar so that the portion of the second clip exposed from the top side of the encapsulant comprises the upper surface of the second attachment region.

16. An electronic device, comprising: a first electronic component coupled to a first side of a substrate, the first electronic component comprising a first component terminal and a second component terminal located on a first major side of the first electronic component; a clip structure comprising: a first clip coupled to the first component terminal; a second clip coupled to a first terminal of the substrate; a connector first portion coupled to the first clip; and a connector second portion coupled to the and separated from the connector first portion by a gap; and an encapsulant covering the first electronic component, portions of the substrate, and portions of the clip structure; wherein: a top side of the first clip and a top side of the second clip are exposed from a top side of the encapsulant.

17. The electronic device of claim 16 further comprising: a second component coupled to the substrate, the second component comprising a first component terminal located on a first major side of the second component; wherein: the second clip is coupled to the first component terminal of the second component.

18. The electronic device of claim 16, wherein: the first clip comprises is coupled to a second terminal of the substrate.

19. The electronic device of claim 16, wherein: the second clip is coupled to the second component terminal of the first electronic component.

20. The electronic device of claim 16, wherein: the gap comprises a groove in the top side of the encapsulant.
